

Article

Two-Dimensional Real-Time Direction-Finding System for UAV RF Signals Based on Uniform Circular Array and MUSIC-WAA

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Abstract: To address the growing security risks posed by unauthorized unmanned aerial vehicle (UAV) activities, this paper proposes a real-time two-dimensional direction-finding (DF) system for UAVs based on radio frequency (RF) signals. This system employs a six-element uniform circular array (UCA), synchronized HackRF One receivers, and a hybrid algorithm integrating the multiple signal classification (MUSIC) method with a novel weighted average algorithm (WAA). By optimizing the MUSIC spectrum search process, the WAA reduces the computational complexity by over 99.9% at a resolution of 0.1° (from 3,240,000 to 1200 spectral function calculations), enabling real-time estimation of the azimuth and elevation angles. The experimental results demonstrate an average azimuth error of 7.0° and elevation error of 7.7° for UAV hovering distances of 30–200 m and heights of 20–90 m. Real-time flight tracking further validates the system's dynamic monitoring capabilities. The hardware platform, featuring omnidirectional coverage ($0\text{--}360^\circ$ azimuth, $0\text{--}90^\circ$ elevation) and dual-band operation (2.4 GHz/5.8 GHz), offers scalability and cost-effectiveness for low-altitude security applications. Despite limitations in the elevation sensitivity due to the UCA's geometry, this work establishes a practical foundation for UAV monitoring, emphasizing computational efficiency, real-time performance, and adaptability to dynamic environments.

Keywords: UAV monitoring; direction-finding; MUSIC algorithm; weighted average algorithm; uniform circular array; real-time tracking; radio frequency



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1. Introduction

Unmanned aerial vehicles (UAVs), as a type of unmanned aircraft, have now developed into high-tech products with highly intelligent and diversified functions. Their applications span diverse fields, such as agricultural production [1–3], the logistics industry [4], and film production [5,6]. With their rapid response and flexible maneuverability, UAVs provide efficient and convenient solutions for various industries, profoundly impacting many aspects of modern society. UAVs continue to expand their application boundaries, showing broad prospects for development.

However, the rapid proliferation of UAV technology has introduced significant challenges to public safety and privacy. Malicious actors are increasingly exploiting UAVs for unauthorized surveillance and airspace intrusions near critical facilities and raising concerns about privacy breaches and airspace security [7]. UAVs have been used to take

unauthorized aerial photographs of private properties, government buildings, and military installations, posing a serious threat to individual privacy and national security [8].

These threats underscore the urgent need for robust UAV monitoring systems. UAV monitoring encompasses detection, identification, and localization, where direction-of-arrival (DOA) estimation serves as a key component of localization and countermeasure deployment. Accurate DOA estimation is crucial for determining the position and trajectory of UAVs, enabling timely interventions to mitigate potential risks. The current direction-finding (DF) methods rely on acoustic, radar, radio frequency (RF), and visual technologies [9–12]. Among these, RF-based systems strike a balance between the operational range, cost-effectiveness, and adaptability to dynamic environments [13]. The RF signals emitted by UAV controllers or onboard communication modules provide a reliable source for passive detection, making them particularly suitable for covert monitoring scenarios [14].

Recent advancements in RF-based DF methods highlight both progress and limitations. For instance, in [15], the authors proposed a UAV direction-positioning method leveraging sparse denoising autoencoders (SDAEs) and deep neural networks (DNNs). This approach enables implementation without requiring phase synchronization, antenna calibration, or a detailed analysis of the antenna's radiation patterns and can be executed using a single-channel RF receiver. The experimental results show that the proposed method achieves a 45° resolution in its estimation on the UAV's azimuth. However, this study was not able to estimate the elevation, and the angle classification direction-finding method based on neural networks exhibited significant limitations in its direction-finding accuracy.

Compared with the angle classification method based on a neural network, a direction-finding method based on array signal processing can achieve a higher estimation accuracy. In [16], the authors utilized a UAV remote controller as the FHSS signal source. The signal is received through a four-element uniform linear array (ULA) and preprocessed using a time–frequency analysis and reconstruction. The MUSIC algorithm and the root-MUSIC algorithm are used to estimate the DOA of the signal. Comparison tests prove that the proposed method significantly improves the accuracy of DF, and the average error in DOA estimation is 1.39° . Similarly, in the research in [17], the authors utilized a software-defined radio (SDR) platform along with a four-element uniform linear array. They processed the signals received by the four phase-coherent RF channels using the MUSIC algorithm. Their experiment results showed that in a static scene, the average estimation error was 1.15° . However, ULA-based systems are fundamentally limited to the azimuth range and lack elevation estimation capabilities. For example, the article in [18] achieved a narrow DOA range of -60° to 60° using a four-element ULA, reporting a 1.9° average azimuth error but no elevation estimations.

In paper [19], the authors attempted to achieve 0 – 360° azimuth detection by utilizing a uniform circular array (UCA). They proposed a USRP-based UAV positioning system that used a five-element UCA antenna and the MUSIC algorithm to determine the azimuth DOA of the UAV. Compared to the actual DOA calculated from the UAV log file, the average error was 18.65° . However, an estimate of the elevation angle was not realized in this study. While these studies have excelled in their accuracy for the azimuth, their inability to resolve the elevation angles or provide omnidirectional coverage restricts their applicability to dynamic, two-dimensional UAV tracking.

A critical gap persists in the existing research: while UCA configurations theoretically enable omnidirectional coverage, practical implementations struggle with sensitivity to the elevation angle and computational inefficiency. The traditional MUSIC-based approaches suffer from exhaustive spectral searches, rendering real-time applications infeasible. For instance, when achieving a resolution of 0.1° , employing a UCA for two-dimensional angle

estimation using the MUSIC algorithm necessitates approximately 3.24 million spectral function computations, limiting the dynamic tracking capabilities. Furthermore, most studies have focused solely on azimuth estimation, neglecting the vertical dimension essential for comprehensive UAV monitoring in complex 3D environments.

Considering the aforementioned limitations, we propose a DF system based on a UCA combined with the WAA [20] and the MUSIC algorithm [21]. Unlike previous methods that have used a ULA, our UCA configuration enables omnidirectional signal reception. Moreover, we achieve real-time estimations of the azimuth and elevation angles of the UAV, addressing the shortcomings of existing 2D DOA estimations. Additionally, integrating the WAA optimizes the MUSIC spectrum, significantly improving the accuracy of DF and the computational efficiency. The main contributions of this research can be summarized as follows:

- A novel DF system is proposed by integrating a UCA with the MUSIC-WAA algorithm. At a resolution of 0.1° , this hybrid approach optimizes the MUSIC spectrum search process, reducing the computational complexity by more than 99.9% compared with that of spectral traversal (from 3,240,000 to 1200 spectral function calculations) while achieving real-time azimuth and elevation estimation.
- A scalable and cost-effective hardware platform is developed using six HackRF One R10 software-defined radio devices, synchronized via a synchronization clock and trigger modules. The system supports omnidirectional coverage ($0\text{--}360^\circ$ azimuth, $0\text{--}90^\circ$ elevation) and dual-band operation (2.4 GHz and 5.8 GHz) by replacing the antenna array;
- Through the UAV hovering experiment (30–200 m distance, 20–90 m altitude), we demonstrate the DF system's accuracy, with average azimuth and elevation errors of 7.0° and 7.7° , respectively. By comparing the real-time data, the effective tracking ability of the DF system for UAVs is verified.

Experimental validation demonstrates an average azimuth error of 7.0° and elevation error of 7.7° for UAVs at 30–200 m distances and 20–90 m altitudes. Real-time flight tracking further confirms the system's dynamic monitoring capabilities. Our system demonstrates robust performance in real-world environments, making it a practical and reliable solution for UAV monitoring. This work bridges theoretical advancements and practical deployments, offering a robust foundation for low-altitude security applications.

The rest of this paper is structured as follows: Section 2 details the proposed method combining the MUSIC algorithm with the WAA; Section 3 describes the DF system's hardware and software components; Section 4 presents the measurement results and analysis; finally, Section 5 concludes this paper with a summary and directions for future work.

2. The Proposed Method

In this section, we explore the application of a swarm intelligence optimization algorithm, the WAA, to optimizing the MUSIC spectrum function for DOA estimation. The MUSIC algorithm is known for its high-resolution DOA estimation capabilities, but its computational inefficiency in the spectrum search process can limit its real-time application. To address this issue, the WAA is integrated with the MUSIC algorithm to improve the optimization efficiency and reduce the computational time.

2.1. The MUSIC Algorithm Based on a UCA

As illustrated in Figure 1, this is a uniform circular array with M elements, where the radius of the array is R and its center serves as the reference point. We assume a far-field

narrowband signal $\tilde{s}(t)$ with a carrier frequency f_0 arriving at the UCA from the direction (θ, φ) . The signal $\tilde{s}(t)$ can be expressed in its complex form as follows:

$$\tilde{s}(t) = s(t)e^{j2\pi f_0 t} \quad (1)$$

where $\tilde{s}(t)$ denotes the complex envelope of $s(t)$, which exhibits a slow temporal variation.

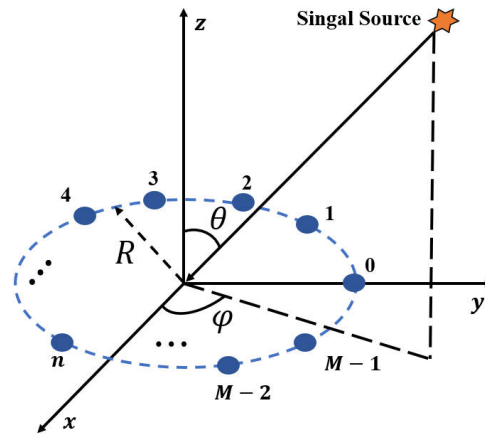


Figure 1. The structure of the UCA.

We define the propagation delay for each array element m with respect to the reference point as τ_m , for $m = 0, 1, \dots, M-1$. If the signal received at the reference point is $\tilde{x}(t)$, the signal received by the m -th element of the array is

$$\tilde{x}_m(t) = \tilde{x}(t - \tau_m) = r(t - \tau_m)e^{j2\pi f_0(t - \tau_m)} \approx s(t)e^{j2\pi f_0(t - \tau_m)} \quad (2)$$

Let $\gamma_m = \frac{2\pi m}{M}$ and $\beta = \frac{2\pi}{\lambda}$, where λ is the wavelength. The position of the m -th array element is

$$\mathbf{p}_m = [R \cos(\gamma_m), R \sin(\gamma_m), 0] \quad (3)$$

and the unit vector pointing in the direction of arrival of the signal is

$$\mathbf{r} = [-\sin \theta \cos \varphi, -\sin \theta \sin \varphi, \cos \theta] \quad (4)$$

Thus, the propagation delay τ_m for the m -th array element relative to the reference point is given by

$$\tau_m = -\frac{R}{c} \sin \theta \cos(\varphi - \gamma_m) \quad (5)$$

where c is the speed of light, and the corresponding phase shift is

$$\phi_m(\theta, \varphi) = -\beta R \sin \theta \cos(\varphi - \gamma_m) \quad (6)$$

The steering vector for the direction of arrival (DOA) is defined as

$$\mathbf{a}(\theta, \varphi) = [e^{-j\phi_0(\theta, \varphi)} e^{-j\phi_1(\theta, \varphi)} \dots e^{-j\phi_{M-1}(\theta, \varphi)}]^T \quad (7)$$

For K narrowband signals $s_1(t), s_2(t), \dots, s_K(t)$ arriving at the UCA with the corresponding DOAs $(\theta_1, \varphi_1), (\theta_2, \varphi_2), \dots, (\theta_K, \varphi_K)$, the baseband signal received at the array can be expressed as

$$\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t) + \mathbf{v}(t) \quad (8)$$

where $\mathbf{A}$ is the array response matrix, and $\mathbf{n}$ denotes the additive white Gaussian noise. The discrete-time complex baseband signal associated with Equation (9) can be given in

$$\mathbf{x}(k) = \mathbf{A}\mathbf{s}(k) + \mathbf{n}(k) \quad (9)$$

The MUSIC algorithm relies on eigenvalue decomposition of the covariance matrix $\mathbf{R}$, which is defined as

$$\mathbf{R} = \frac{1}{N} \sum_{k=1}^N \mathbf{x}(k)\mathbf{x}^H(k) \quad (10)$$

The following formula can be obtained through matrix decomposition of the matrix $\mathbf{R}$:

$$\mathbf{R} = \mathbf{E}\mathbf{\Sigma}\mathbf{E}^H \quad (11)$$

where $\mathbf{E}$ is the eigenvector of the matrix $\mathbf{R}$, and $\mathbf{\Sigma}$ is the eigenvalue of the matrix $\mathbf{R}$.

According to the size order of the eigenvalues, the eigenvector corresponding to the largest eigenvalue $\mathbf{\Sigma}_S$ equal to the number of signals K is regarded as the signal subspace $\mathbf{U}_S$, and the eigenvector corresponding to the remaining $M - K$ eigenvalues $\mathbf{\Sigma}_N$ is regarded as the noise subspace $\mathbf{U}_N$. Equation (12) is obtained, as shown below:

$$\mathbf{R} = \mathbf{U}_S\mathbf{\Sigma}_S\mathbf{U}_S^H + \mathbf{U}_N\mathbf{\Sigma}_N\mathbf{U}_N^H \quad (12)$$

After decomposing the matrix $\mathbf{R}$, the MUSIC pseudospectrum is defined as

$$P_{MUSIC}(\theta, \varphi) = \frac{1}{\mathbf{a}^H(\theta, \varphi)\mathbf{U}_N\mathbf{U}_N^H\mathbf{a}(\theta, \varphi)} \quad (13)$$

The peaks in this spectrum correspond to the directions of arrival of the received signals.

2.2. The Weighted Average Algorithm

The weighted average algorithm is a novel metaheuristic optimization technique that aims to balance exploration and exploitation. The core idea behind this algorithm is to iteratively adjust the search parameters and calculate the weighted average position. This process helps guide the optimization and makes the WAA suitable for solving complex optimization problems in various domains, including DOA estimation. The optimization process involves the following key steps:

(1) The initialization phase

This phase randomly generates an initial set of candidate solutions and calculates the fitness value for each candidate solution. The set of candidate solutions is denoted as the matrix $\mathbf{X}$, which is defined using Equation (14) and randomly generated within the predefined search space according to Equation (15).

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & x_{2,1} & \dots & x_{1,n} \\ x_{1,2} & x_{2,2} & \dots & x_{1,n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N,1} & x_{N,2} & \dots & x_{N,n} \end{bmatrix} \quad (14)$$

where $x_{i,j}$ denotes the position of the i -th solution within the j -th dimension, N denotes the total number of candidate solutions, and n denotes the dimensionality of the problem.

$$x_{ij} = rand \cdot (UB_j - LB_j) + LB_j, \quad i = 1, 2, \dots, N; \quad j = 1, 2, \dots, n \quad (15)$$

where $rand$ is a random number between 0 and 1, LB_j denotes the j -th lower bound value, and UB_j denotes the j -th upper bound value of the given problem.

(2) The weighted average position

To determine the weighted average position, the initial step involves evaluating the fitness of each participant. Next, the candidate solutions are re-organized based on one of two optimization goals: the larger the better (LTB) or the smaller the better (STB). If LTB, we sort the candidate solutions from larger to smaller fitness values, while if STB, we sort the candidate solutions from smaller to larger fitness values. Following this, the first N_C candidates from the set of candidate solutions are chosen to calculate the weighted average position, utilizing the equations provided below:

$$N_C = (nP - 4) \frac{iter - 1}{1 - Max_I} + nP \quad (16)$$

$$SumF = \sum_{i=1}^{N_C} F(X_i) \quad (17)$$

$$X_{Miu} = \frac{\sum_{i=1}^{N_C} X_i \cdot (SumF - F(X_i))}{SumF(N_C - 1)} \quad (STB) \quad (18)$$

$$X_{Miu} = \frac{\sum_{i=1}^{N_C} X_i \cdot F(X_i)}{SumF} \quad (LTB) \quad (19)$$

where nP is the number of populations, X_i is the i -th candidate solution, F is the function used to calculate the fitness value, $SumF$ is the total of all of the fitness values from the chosen candidates, and X_{Miu} is the weighted average position. Additionally, $iter$ is the current iteration number, and Max_I is the maximum iteration limit. N_C is the number of selected candidate solutions. Notably, Equations (18) and (19) correspond to the weighted average position equations when choosing the optimization goals of STB or LTB, respectively.

(3) Defining the search phase: exploration or exploitation

For each candidate solution in the population, it is decided whether to explore or exploit according to Equation (20).

$$k_1 = (\alpha \cdot rand - 1) \sin\left(\frac{\pi \cdot iter}{Max_{Iter}}\right) \quad (20)$$

where $0 \leq k_1 \leq 1$, and if $k_1 < 0.5$, the candidate solution position update progresses to the exploration phase; otherwise, it progresses to the exploitation phase; $rand$ is a random value ranging from 0 to 1; and α is an adjustable constant to control the balance between the exploration and exploitation phases.

(4) The exploitation phase

The exploitation strategy simulates how the population of search agents moves towards the search spaces with a high probability of exploiting new global best values. There are three movement strategies employed during the exploitation phase, which are detailed in the following equations. Each of these formulas represents a distinct exploitation strategy.

$$X_i(iter + 1) = w_{11} \cdot (X_{Miu}(iter) - X_{GBest}(iter)) + w_{12} \cdot (X_{Miu}(iter) - X_{PBest}(iter)) + w_{13} \cdot X_{Miu}(iter) \quad (21)$$

$$X_i(iter + 1) = w_{21} \cdot (X_{Miu}(iter) - X_{PBest}(iter)) + w_{22} \cdot X_{PBest}(iter) \quad (22)$$

$$X_i(iter + 1) = w_{31} \cdot (X_{Miu}(iter) - X_{GBest}(iter)) + w_{32} \cdot X_{GBest}(iter) \quad (23)$$

where $w_{ij}, i = 1, 2, 3; j = 1, 2, 3$ are random values ranging from 0 to 1, and X_{pBest} and X_{GBest} represent the personal best and global best position in the $iter$ iteration numbers, respectively.

The selection between the three distinct movement strategies is controlled by $k_2 \in \{1, 2, 3\}$, applying Equations (21)/(22)/(23), respectively.

(5) The exploration phase

The exploration phase aims to identify novel potential solutions across the global domain, thereby enhancing the population diversity and avoiding falling into local optima. There are two different exploration strategies during the exploration phase. The first strategy is based on the Lévy's flight model, which is defined by the following equations:

$$S = \frac{U}{|V|^{\frac{1}{\beta}}} \quad (24)$$

$$U = \text{normal}(0, \sigma_u^2) \quad (25)$$

$$V = \text{normal}(0, \sigma_v^2) \quad (26)$$

$$\sigma_u = \left\{ \frac{\Gamma(1 + \beta) \sin\left(\frac{\pi\beta}{2}\right)}{\Gamma\left(\frac{1+\beta}{2}\right) \beta 2^{\frac{\beta-1}{2}}} \right\}^{\frac{1}{\beta}} \quad (27)$$

$$X_{i,j}(iter + 1) = X_{GBest_j}(iter) + S \quad (28)$$

where Γ represents the Gamma distribution function, while S is the step length of the Lévy flight, which is influenced by the parameter β . Additionally, U and V satisfy normal distributions with standard deviations equal to σ_u and σ_v , respectively, and both have a mean of 0. During the Lévy flight process, the step length is determined by the value of β . The notation X_{GBest_j} is the j -th position in the global best solution at the iteration $iter$, whereas $X_{i,j}(iter + 1)$ denotes the j -th position of the i -th solution at the iteration $iter + 1$.

The second exploration strategy is defined by the following equation:

$$X_i(iter + 1) = \text{rand} \cdot (UB_{min} - LB_{min}) + LB_{min} \quad (29)$$

where LB_{min} and UB_{min} represent the lowest values of the lower and upper bounds across all dimensions, respectively.

The selection between the two distinct movement strategies is controlled by the variable k_3 , where $0 \leq k_3 \leq 1$. If $k_3 > 0.5$, Equation (28) is employed to update the candidate solution position; otherwise, Equation (29) is applied.

To illustrate the optimization process better, a flowchart summarizing the steps is shown in Figure 2. It can be seen that the main process of the WAA includes three parts: initialization, recording the optimal solution, and updating the population. The core aspect lies in the calculation of the weighted average position and the balance between the exploration and exploitation phases. The exploration phase enhances the global search ability through two exploration strategies, while the exploitation phase improves the local optimization ability through three exploitation strategies. After several iterations, the WAA can find the precise optimal solution for the objective function.

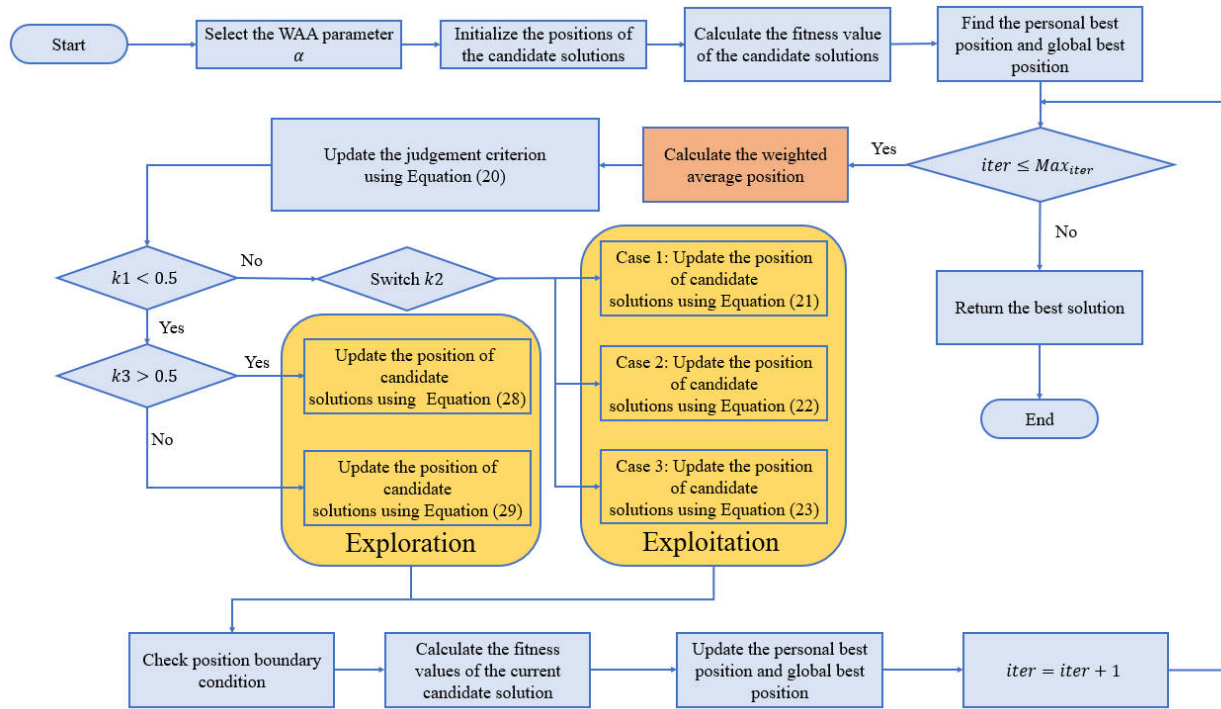


Figure 2. Flowchart of the WAA.

2.3. Simulation of Optimization of the MUSIC Spectrum Function Based on the WAA

We investigate the DOA techniques for a single UAV. In this context, the MUSIC spectrum function exhibits its peak in the direction of arrival of the source of RF signals for the UAV. Using this characteristic, we reformulate the DF problem for a single UAV as an optimization problem, specifically focusing on identifying the maximum value of the MUSIC spectrum function to determine the DOA of the corresponding UAV RF signal. To convert this optimization problem into a minimization problem (usually, optimization algorithms are used to solve the minimum optimization), we multiply the MUSIC spectrum function by -1 . Thus, the optimization problem becomes

$$\min f(x_1, x_2) = -P_{MUSIC}(x_1, x_2) = -\frac{1}{\mathbf{a}^H(x_1, x_2)\mathbf{U}_N\mathbf{U}_N^H\mathbf{a}(x_1, x_2)} \quad (30)$$

subject to

$$\begin{cases} 0^\circ \leq x_1 \leq 90^\circ \\ 0^\circ \leq x_2 < 360^\circ \end{cases} \quad (31)$$

where x_1 and x_2 denote the elevation and azimuth angles, respectively, in the MUSIC spectral function.

To verify the optimization effect of the WAA on the MUSIC spectrum function, the simulation experiments were carried out by MATLAB 2022b. In the experiment, four incident signals with different angles of arrival were set. The frequency of all of the signals was 2.4 GHz, and the signal-to-noise ratio (SNR) for all of the signals was 5 dB. The receiving array was a 6-element UCA with a radius of 6.25 cm. The population size of the WAA was set to 30, the number of iterations was set to 40, and the WAA parameter $\alpha = 10$. The angle values (elevation, azimuth) of the four test signals, along with the results of the WAA optimization and the corresponding absolute errors, are presented in Table 1. These results demonstrate that the absolute errors within 1° in the WAA indicate excellent precision in estimating the DOA.

Table 1. A comparison between the incidence angle and optimized result from the WAA.

Index	a	b	c	d
Incidence Angle	(20.0°, 60.0°)	(40.0°, 100.0°)	(50.0°, 160.0°)	(70.0°, 240.0°)
WAA Optimization Result	(19.9°, 60.6°)	(40.1°, 100.2°)	(50.1°, 159.4°)	(70.1°, 240.4°)
Absolute Error	(0.1°, 0.6°)	(0.1°, 0.2°)	(0.1°, 0.6°)	(0.1°, 0.4°)

To assess the performance of the WAA, we compare it with other popular optimization algorithms, such as the Whale Optimization Algorithm (WOA) [22], the Sparrow Search Algorithm (SSA) [23], the Particle Swarm Optimization Algorithm (PSO) [24], and the Grey Wolf Optimization Algorithm (GWO) [25]. All of the algorithms are tested under the same simulation conditions, with identical population sizes and iteration numbers. The optimal fitness curves for different optimization algorithms across iterations are shown in Figure 3.

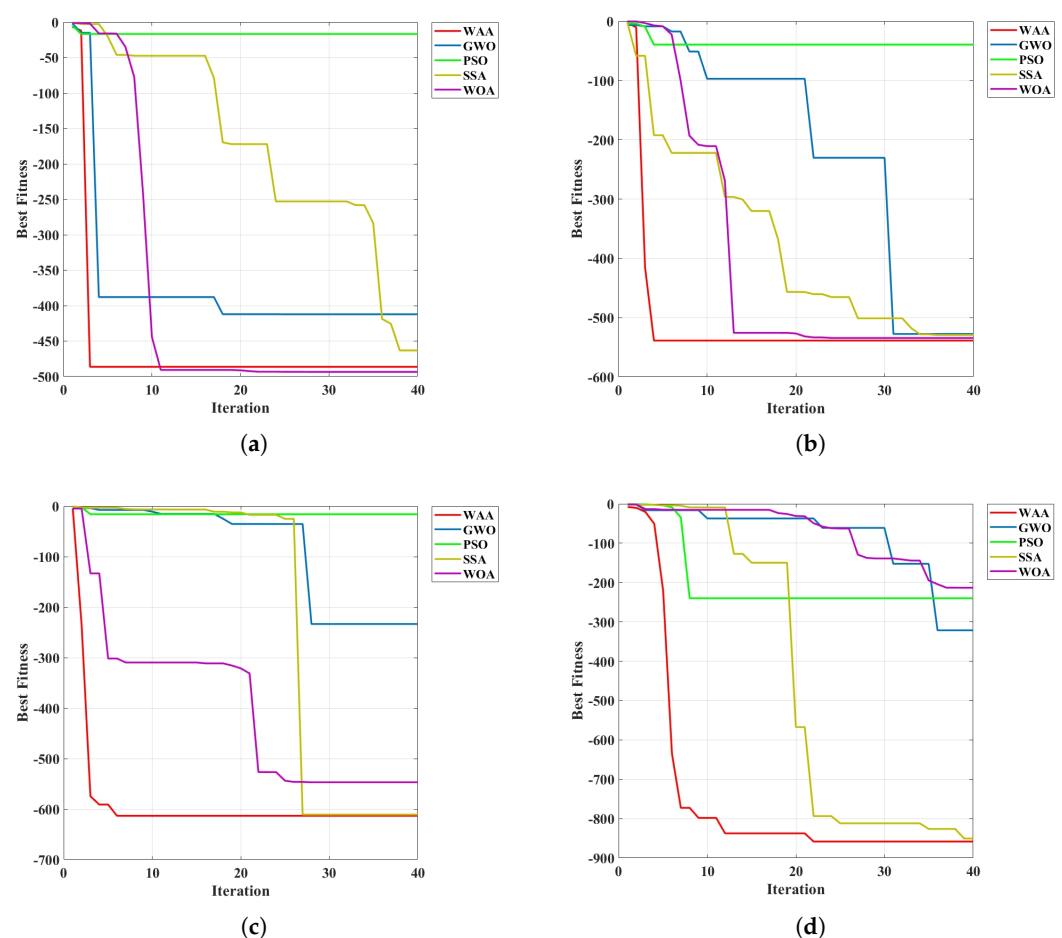


Figure 3. Comparison of optimal fitness curves across different optimization algorithms. (a) Simulation experiment of incident angle (20.0°, 60.0°). (b) Simulation experiment of incident angle (40.0°, 100.0°). (c) Simulation experiment of incident angle (50.0°, 160.0°). (d) Simulation experiment of incident angle (70.0°, 240.0°).

The WAA significantly outperforms the other optimization algorithms, converging to the optimal solution in fewer iterations. Specifically, the WAA completed the optimization within just 40 iterations (where the MUSIC spectrum function performed 1200 evaluations during the 40 iterations), reducing the computation time by approximately 99.9% compared to that of the traditional MUSIC spectrum traversal method that searched at every 0.1° increments (where the MUSIC spectrum function was evaluated 3,240,000 times). In summary, the WAA significantly enhances the computational efficiency of the MUSIC spectral

function, exhibits an outstanding performance in terms of its processing speed, and ensures the feasibility of real-time DOA estimation.

3. Description of the DF System

The proposed DF system, as illustrated in Figure 4, comprises five core components: a UCA antenna array, a HackRF One [26] receiver group, a synchronous clock module, a synchronous trigger module, and a host controller. These components work in concert to enable the real-time reception and processing of UAV RF signals, allowing for the accurate estimation of the DOA and tracking of a UAV's movement.

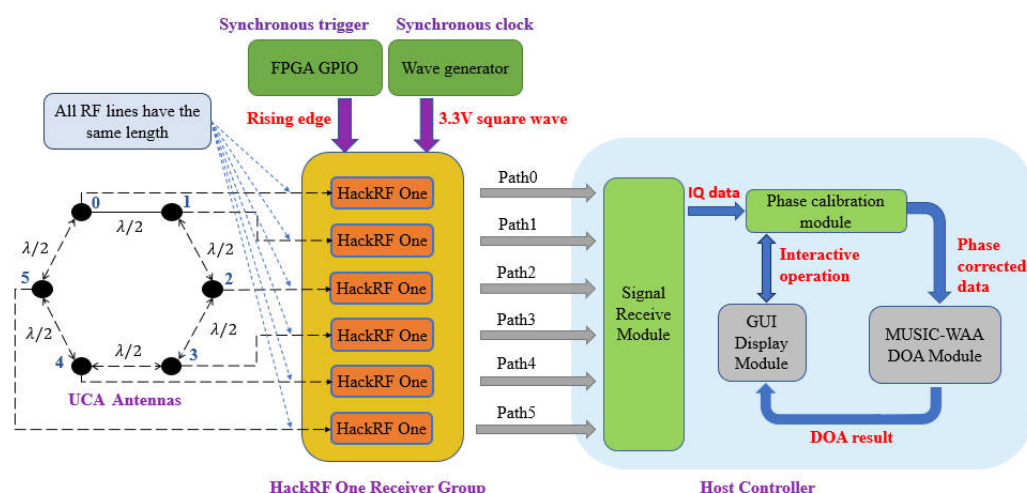


Figure 4. Block diagram of the DF system.

The UAV RF signal is received through the UCA antenna units and transmitted to the corresponding HackRF One unit via RF lines of the same length. The group of HackRF One receivers achieves synchronous signal reception through the synchronous clock and trigger modules. The host controller performs phase calibration on the received multi-channel signals, followed by real-time calculations, with the results displayed via the GUI module.

(1) UCA antennas

The UCA antennas are the system's primary signal reception component. They consist of six 2.4 GHz folded dipole antennas arranged in a circular configuration with a radius of 6.25 cm (corresponding to half-wavelength spacing for 2.4 GHz signals, which have a wavelength of 12.5 cm). The antenna array operates within the 2.4 GHz frequency band, providing a gain of 6 dBi per antenna. This configuration enables omnidirectional reception of RF signals while maintaining phase coherence across all of the antenna elements, which is crucial for accurate DOA estimation.

(2) The HackRF One Receiver Group

The HackRF One receiver group comprises six individual HackRF One R10 software-defined radio (SDR) devices, with each connected to a corresponding antenna element in the UCA. The HackRF One is a versatile SDR that supports a frequency range of 1 MHz to 6 GHz, with a maximum analog-to-digital conversion (ADC) sampling rate of 20 MSPS. This wide frequency range makes it well suited to receiving UAV communication signals, which typically operate in the 2.4 GHz and 5.8 GHz bands. To ensure synchronized operation of the receiver devices, a common clock source is shared across all HackRF One units. This synchronization is essential for accurate phase calibration and signal processing, as phase discrepancies between channels could otherwise lead to inaccurate DOA estimations.

(3) The Synchronous Clock Module

The synchronization clock module is an eight-channel, 3.3V, 10 MHz square wave generator (BG7TBL) [27] that provides a unified reference clock for all HackRF One units.

This ensures that all of the devices operate at the same sampling rate, allowing for consistent data acquisition across the receiver channels.

(4) The Synchronous Trigger Module

We modified the source code of the HackRF One so that it waited for a rising-edge trigger signal after it was turned on, and when the trigger signal was received, the HackRF One started the signal sampling. The synchronization trigger module was implemented using an Altera EP4CE10E22 FPGA core board [28]. This module generates a 5-pulse-per-second (PPS) rising-edge signal through GPIO ports to synchronize the timing of the data acquisition across all SDR devices. The synchronization trigger ensures that all signals are captured at precisely the same moment, thus eliminating timing mismatches between channels.

(5) The Host Controller

The host controller integrates several functional modules developed using the GNU Radio framework v3.10.7.0, including GNU Radio C++ scripts and the GNU Radio Companion graphical environment, facilitating the real-time reception and processing of signals. Four key modules are implemented on the host controller. The signal receive module collects the raw in-phase and quadrature (IQ) data streams from the HackRF One receivers. The phase calibration module compensates for the phase discrepancies between different channels by referencing a calibration signal. The MUSIC-WAA DOA module implements the MUSIC algorithm for DOA estimation, further optimized by the WAA to enhance the accuracy and computational efficiency of the system. The WAA's optimization reduces the computational burden typically associated with MUSIC spectrum traversal, enabling real-time processing; the GUI display module provides a user interface for real-time monitoring of the system's performance. It comprises the iteration curve of the MUSIC spectrum function optimized by the WAA, the DOA estimation results, the azimuth and elevation planes of the MUSIC spectrum function corresponding to the DOA estimation results, the signal waveforms received by each antenna unit, the phase differences between each channel signal and the reference signal of channel 0, the amplifier gain configuration module, and the phase calibration setup module. The system's GUI is shown in Figure 5.

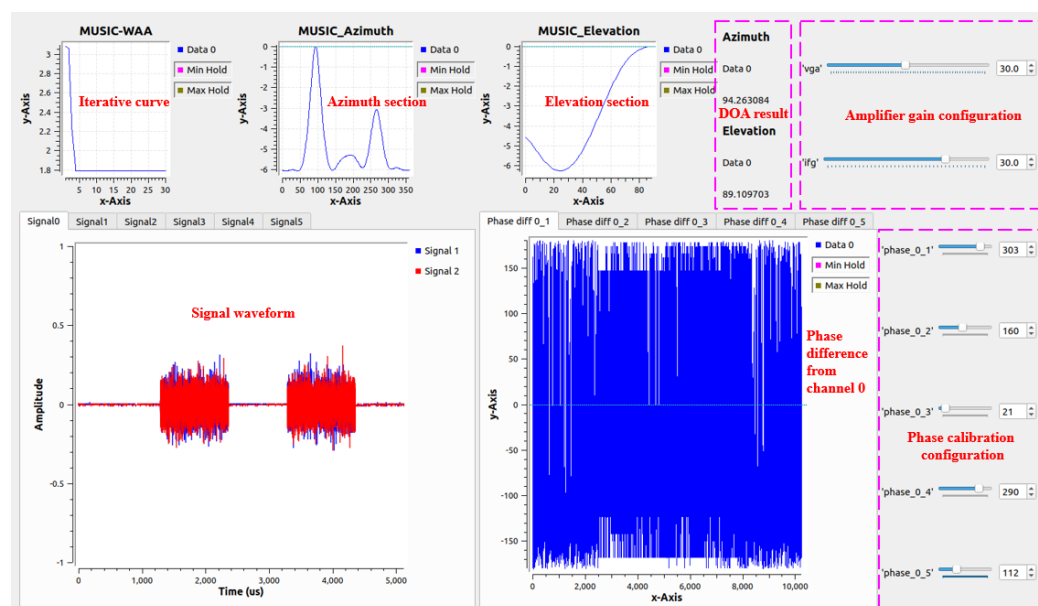


Figure 5. The system's GUI.

The physical components of the system are shown in Figure 6. The UCA antennas are mounted onto a tripod at a height of 120 cm. The HackRF One receiver group receives

the signal from the corresponding antenna. The FPGA-based synchronization modules ensure precise hardware synchronization across the receiver devices, while the GNU Radio software stack facilitates low-latency signal processing. The combined system provides real-time DOA estimation, enabling dynamic UAV tracking during flight.

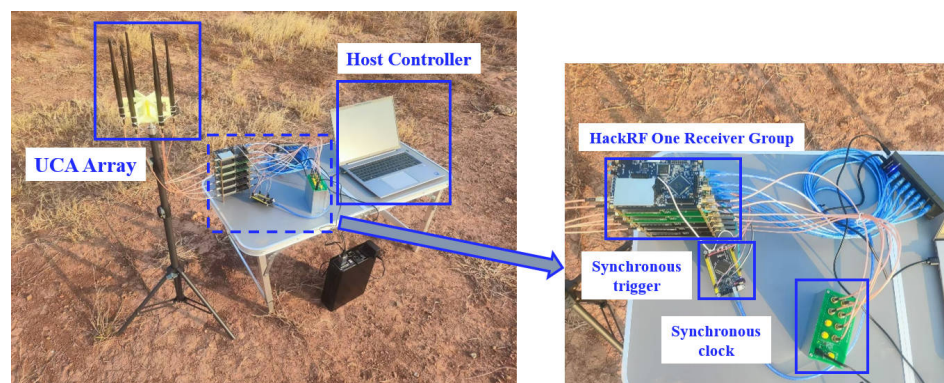


Figure 6. Physical components of the DF system.

4. Experimental Validation and Analysis

This section presents a comprehensive evaluation of the proposed two-dimensional real-time direction-finding system for UAV RF signals. Two primary experimental scenarios were designed: static hovering measurements to evaluate the accuracy of static angle estimation and dynamic flight tracking to assess the real-time performance. Ground-truth data, extracted from the UAV's flight logs, served as a precise reference for the error analysis. The experiments aimed to validate the system's accuracy, computational efficiency, and dynamic tracking capabilities under real-world conditions.

4.1. System Initialization

Before conducting the experiments, the system was initialized to ensure the optimal performance during data collection. Several parameters were configured, taking into account the limited processing throughput of the host computer. The sampling rate of the HackRF One devices was set to 2 MHz, and the center frequency was fixed at 2407 MHz. Additionally, the intermediate frequency gain and the variable-gain amplifier were both set to 30 dB to enhance the sensitivity for signal reception.

It should be noted that the MUSIC algorithm primarily depends on the number of snapshots rather than the sampling rate for its performance. By intentionally setting the sampling rate to 2 MHz and the snapshots to 10,240, this configuration effectively mitigates potential computational bottlenecks caused by an excessive volume of data while maintaining a sufficient signal characterization capability. This optimized sampling parameter selection balances the algorithmic requirements with practical system constraints, ensuring reliable processing without compromising the spatial spectrum estimation accuracy.

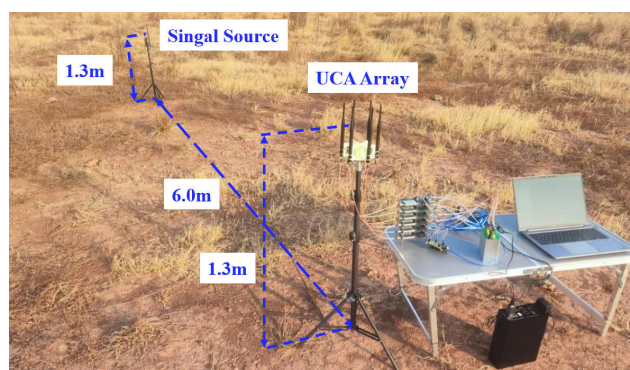
For the MUSIC algorithm, the following parameters were configured: the number of snapshots was set to 10,240; the number of antenna array elements was set to 6; the radius of the array elements was 6.25 cm; and the number of sources was set to 1 to match the single UAV being tracked. These parameters ensured that the MUSIC algorithm could accurately process the signals received from the UAV.

In parallel, the WAA parameters were set as follows: the population size was set to 30, the number of iterations was set to 30, and the WAA parameter α was set to 10. These settings were selected to balance the algorithm's performance and computational efficiency. Table 2 presents a comprehensive list of all of the parameters required for system initialization.

Table 2. Parameter settings for system initialization.

HackRF One	Parameter	Sample rate	Center frequency	Intermediate frequency gain	Variable-gain amplifier	-
	Value	2 MHz	2407 MHz	30 dB	30 dB	-
MUSIC	Parameter	Snapshots	Antenna number	UCA radius	Signal wavelength	Source number
	Value	10,240 MHz	6	6.25 cm	12.5 cm	1
WAA	Parameter	Iteration number	α	-	-	-
	Value	30	10	-	-	-

Once the system parameters were configured, a phase calibration procedure was performed. It is worth noting that the gain, phase response, and group delay of each channel are different due to the manufacturing tolerance of the RF front-end components of the different channels during system startup, and these differences change after each system restart. Therefore, channel calibration is required after each system startup to synchronize all of the receiver channels, ensuring accurate signal processing. A continuous radio wave signal at 2.407 GHz was used as the calibration source. The calibration signal was positioned 6 m to the east of the UCA, aligned with the 0-channel antenna, ensuring that the antenna array was in the far-field region and met the necessary conditions for the calibration. As depicted in Figure 7, this process established a baseline for phase synchronization across all of the receiver channels.

**Figure 7.** Calibration process of the DF system.

The calibration process adjusted the phase difference between each channel to the correct value. Since the calibration signal is located in the same plane as the UCA, the wave path difference between the signals received from channels 1, 2, 3, 4, and 5 relative to channel 0 should be $\frac{\lambda_0}{4}$, $\frac{3\lambda_0}{4}$, λ_0 , $\frac{3\lambda_0}{4}$, and $\frac{\lambda_0}{4}$, respectively, and the corresponding phase difference is 90° , 270° , 360° , 270° , and 90° . The phase discrepancies between the channels were compensated for by the phase calibration module in the host computer, thereby ensuring that the system was ready for accurate direction-finding measurements.

4.2. The Experimental Setup

The experimental site is located in the suburbs with less radio signal interference. The DJI AIR 2S commercial UAV (DJI, Shenzhen, China) [29], selected for its representative RF characteristics and operational versatility, was employed as the target UAV in this study.

The DJI AIR 2S features configurable downlink signals operating within the 2.4 GHz and 5.8 GHz transmission bands. It offers a maximum transmission range of 12 km and a flight duration of approximately 30 min.

In addition, the DJI AIR 2S provides detailed flight logs, including latitude, longitude, and altitude. These logs were essential for calculating the true azimuth and elevation angles, serving as a benchmark for evaluating the performance of the DF system.

For the experiments, the signal characteristics of the UAV were set as follows: the transmission resolution was Hd mode, the frequency band was 2.4 GHz, the channel mode was manual operation, and the bandwidth was 10 MHz, with a center frequency of 2.407 GHz. All of the parameter settings are given in Table 3.

Table 3. Parameter settings for the DJI AIR 2S.

Parameter	Transmission	Frequency Resolution	Channel Band	Bandwidth Mode	Center Frequency
Value	HD mode	2.4 GHz	Manual operation	10 MHz	2407 MHz

The waveform, spectrum, and waterfall diagram of the UAV's RF signal are shown in Figure 8. The UAV's RF signal exhibited discontinuous and irregular signal patterns within the 2.407 GHz band. The signal's bandwidth and center frequency allowed for effective reception and processing by the HackRF One receiver, ensuring compatibility with the HackRF One receiver's configuration.

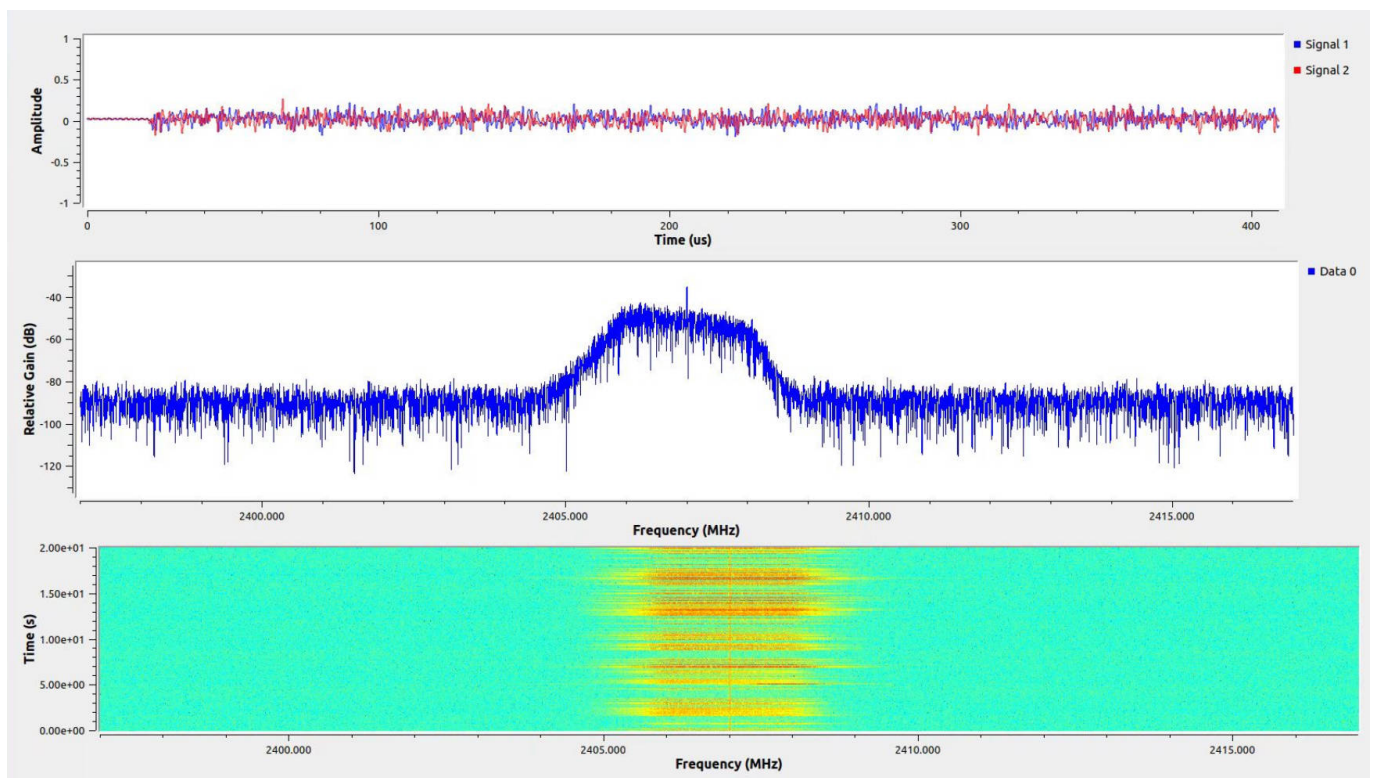


Figure 8. RF signal characteristics of the UAV.

During the experiment, the UAV operated at varying distances, ranging from 30 m to 200 m, from the UCA receiving system and at altitudes between 20 m and 90 m. These parameters provided a comprehensive simulation of typical monitoring scenarios, including both near-field and far-field conditions, as well as low- and medium-altitude scenarios.

The 2D and 3D trace and hovering points of the two flights are shown in Figure 9. There were 26 hovering points from the two flight tests. The red line represents the trace of the first flight experiment, and the green line represents the trace of the second flight experiment. The positions of the two hover flights are indicated by different markers and are distinct from the flight origin markers. The blue circular markers represent the hovering points of the first flight, the purple circular markers represent the hover point of the second flight, and the orange circle represents the takeoff point of the UAV. The azimuth is 0° in the direction of east and increases in the counterclockwise direction, while the elevation angle is 0° in the normal direction of the horizontal plane.

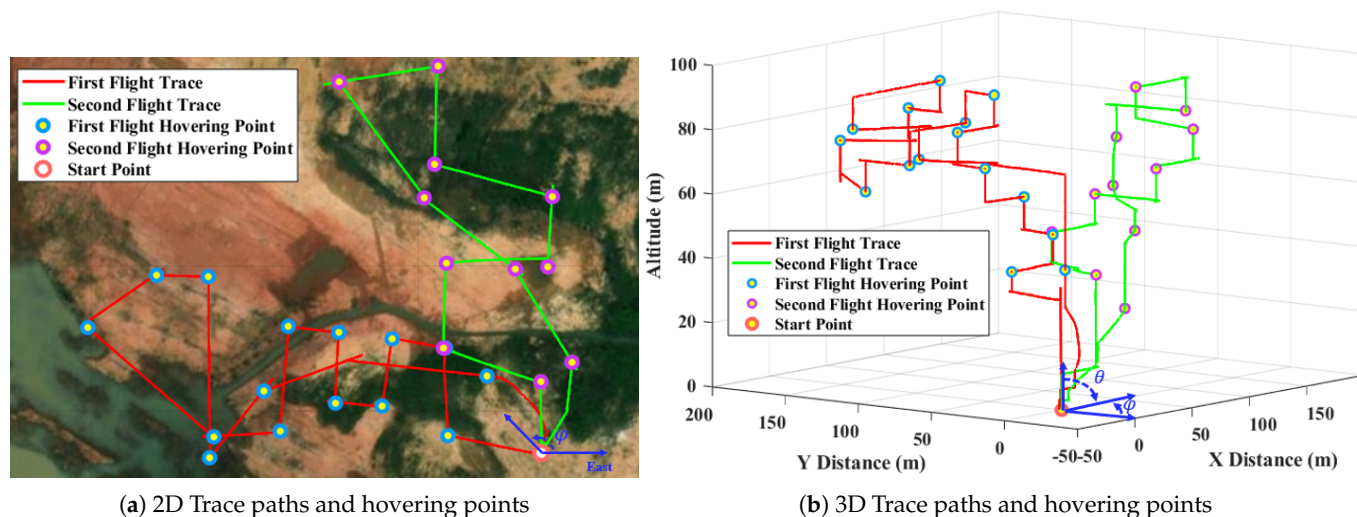


Figure 9. Trace paths and hovering points of the UAV's two flights.

Each flight test involved the UAV hovering at distinct locations, with a minimum hovering duration of 10 s per position. This ensured that sufficient data were collected for accurate angle measurement. Notably, the UAV's takeoff position was fixed at 0.5 m away from the antenna array, and for the purposes of this experiment, the array was considered to be co-located with the UAV's takeoff point. This arrangement enabled consistent and reliable measurements across the flight experiments.

4.3. Analysis of the Angle Measurements of the UAV's Hovering Point

The angle measurements of the UAV's hovering points were obtained by comparing the results of the real-time direction-finding system with the angles calculated from the UAV's flight log. The longitude, latitude, and altitude of each hovering point were extracted from the flight logs, and the distance and height differences from the starting point were calculated, as shown in Table 4. The UAV's hovering positions, indexed from 1 to 26, were grouped into two distinct flight sequences. The first flight sequence consisted of 15 hovering points, while the second flight sequence had 11. Index 0 in Table 1 represents the UAV's takeoff and landing point.

The distance between the 26 hovering points ranged from 30 m to 200 m, and the altitude varied between 30 m and 90 m. These variations in distance and height influenced the direction of arrival of the signal, resulting in changes in the azimuth and elevation angles. The calculation of the azimuth angle is based on Vincenty's formulae [30], a widely adopted geodetic method for computing the distance between two points in an ellipsoidal Earth model by leveraging their geographic coordinates (latitude and longitude). After determining the horizontal distance between the UAV's hovering position and takeoff point, the true azimuth angle of the hovering location is derived by calculating the angle between

the geodesic line connecting these two points and the reference direction of east (defined as 0° , with the angles increasing counterclockwise). Subsequently, the elevation angle is computed through the arctangent function, incorporating the vertical difference in altitude between the two points.

Table 4. Distance and height of UAV hovering points.

Index	Latitude	Longitude	Altitude (m)	Distance (m)	Height (m)
0 (start)	25.76984011	114.7491364	140.4	0.0	0.0
1	25.76990561	114.7487381	180.8	40.6	40.4
2	25.77024192	114.7487277	189.8	60.5	49.4
3	25.77027657	114.7485007	200.1	80.0	59.7
4	25.77001880	114.7484583	210.6	70.8	70.2
5	25.77002978	114.7482601	220.7	90.3	80.3
6	25.77030159	114.7482745	230.4	100.4	90.0
7	25.77032455	114.7480587	220.5	120.6	80.1
8	25.76992283	114.7480253	210.0	111.7	69.6
9	25.76990013	114.7477439	200.5	139.7	60.1
10	25.77032015	114.7472073	210.5	200.5	70.1
11	25.77051998	114.7474993	220.7	180.5	80.3
12	25.77051499	114.7477191	230.4	160.5	90.0
13	25.76982094	114.7477251	220.5	141.5	80.1
14	25.77007673	114.7479557	210.4	121.2	70.0
15	25.77013401	114.7489039	180.5	40.0	40.1
16	25.77011233	114.7491332	180.4	30.2	40.0
17	25.77024124	114.7487190	190.6	60.9	50.2
18	25.77056632	114.7487316	200.1	90.1	59.7
19	25.77055245	114.7491622	210.3	79.1	69.9
20	25.77082168	114.7491817	220.7	109.0	80.3
21	25.77094563	114.7486809	230.3	130.7	89.9
22	25.77132218	114.7486960	220.2	170.0	79.8
23	25.77126209	114.7482753	220.1	179.6	79.7
24	25.77081717	114.7486373	200.4	119.2	60.0
25	25.77054422	114.7490243	190.4	78.9	50.0
26	25.77018674	114.7492656	170.1	40.7	29.7

The measurement results for each hovering point are obtained by averaging the calculation results stored locally by the DF system during the UAV's hovering period after excluding outliers. The identification of outliers is based on the median absolute deviation (MAD) criterion, defined as data points differing by more than three times the median deviation. The measurement results are presented in Table 5, which contains the count of the measurement results, the average value of the results, the percentage of outliers, and the absolute error compared to the angle values calculated from the UAV flight logs.

As indicated in Table 5, the maximum error for the azimuth angles measured across the 26 UAV hovering points was 108.8° , while the minimum error was 0.1° , and the average azimuth measurement error was 10.9° . For the elevation angles, the maximum error was 37.1° , the minimum error was 0.2° , and the average error was 8.9° . A notable outlier was observed at the 11th measurement point, where both the azimuth and elevation errors substantially exceeded the average values more than 10 times. This can be attributed to the extended propagation distance causing signal attenuation, the low SNR resulting in measurement errors, and the severe multipath interference induced by shrubbery and experimental equipment along the transmission path. By combining the real-time direction-finding results for the UAV shown in Figures 10 and 11, it can be seen that the error was

caused by the multipath effect caused by obstacles, while the other test points did not show such significant errors.

Considering that this measurement point represents a special case of multipath interference rather than a systematic error, we excluded this outlier from the subsequent performance analysis to prevent distortion of the system evaluation metrics. After its exclusion, the maximum error for the azimuth angles was 19.7° , the maximum error for the elevation angles was 23.5° , and the average errors for the azimuth and elevation angles were reduced to 7.0° and 7.7° , respectively.

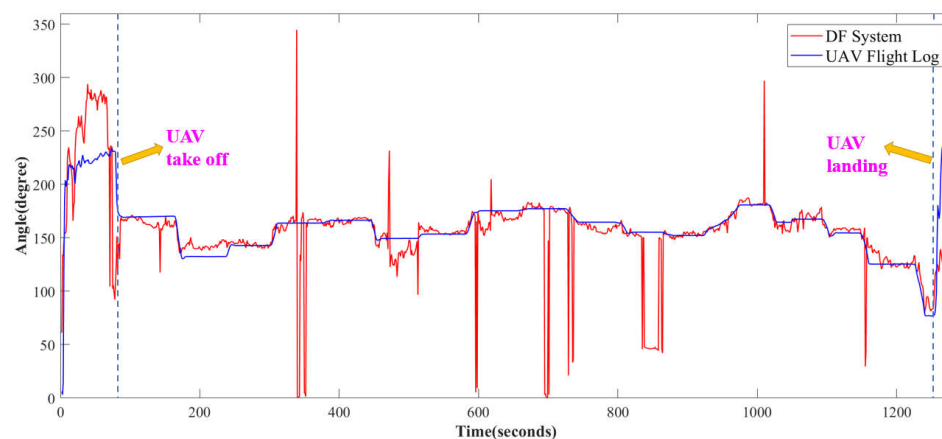
Table 5. Comparison of the measured azimuth and elevation angles from the DF system and calculated values from the UAV flight logs.

Index	UAV Flight Log		DF System				Absolute Error	
	Azimuth ($^\circ$)	Elevation ($^\circ$)	Azimuth ($^\circ$)	Elevation ($^\circ$)	Count	Outliers (%)	Azimuth ($^\circ$)	Elevation ($^\circ$)
0 (start)	-	-	-	-	-	-	-	-
1	169.7	45.1	162.6	36.5	6973	11.09	7.1	8.6
2	132.6	50.7	142.3	44.2	6034	4.21	9.7	6.5
3	142.8	53.3	142.8	57.4	3759	4.52	0.1	4.1
4	163.8	45.2	166.5	42.9	6837	25.82	2.7	2.3
5	166.5	48.4	168.1	35.9	3085	3.95	1.6	12.5
6	149.4	48.1	134.9	24.6	5096	9.52	14.5	23.5
7	153.6	56.4	154.6	60.5	5508	5.52	1.0	4.1
8	175.3	58.1	167.4	60.0	2818	4.05	7.9	1.9
9	177.3	66.7	176.9	72.6	3753	18.55	0.4	5.9
10	164.6	70.7	157.3	81.1	4296	2.89	7.3	10.4
11	155.3	66.0	46.5	28.9	3748	1.70	108.8	37.1
12	155.2	60.7	153.9	77.2	3631	5.48	1.3	16.5
13	180.9	60.5	181.4	57.7	4030	11.34	0.5	2.8
14	167.5	60.0	170.6	79.6	3618	12.24	3.1	19.6
15	125.5	45.0	124.8	61.9	4022	14.00	0.7	16.9
16	90.1	37.0	100.5	30.7	1745	3.04	10.4	6.3
17	133.1	50.5	143.3	50.3	4820	13.86	10.2	0.2
18	116.5	56.5	124.4	55.8	2810	11.10	7.9	0.7
19	88.0	48.5	107.7	38.1	2814	9.70	19.7	10.4
20	87.5	53.6	98.1	51.1	2143	6.07	10.6	2.5
21	110.3	55.5	118.9	65.5	2954	9.31	8.6	10.0
22	105.0	64.9	115.4	69.0	3754	4.05	10.4	4.1
23	118.7	68.7	128.3	66.8	3350	5.04	9.6	1.9
24	114.7	63.2	123.0	64.0	2682	4.88	8.3	0.8
25	98.1	57.6	108.2	50.5	3887	9.67	10.1	7.1
26	71.2	53.8	82.3	39.5	2548	10.68	11.1	14.3
Average			-				10.9	8.9
Average remove11			-				7.0	7.7

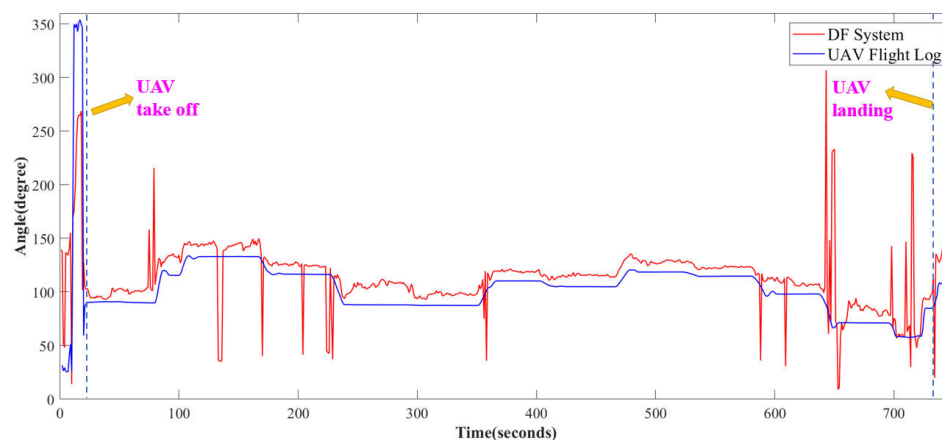
It is evident that the DF system's measurements of both the azimuth and elevation angles exhibit a good match compared to those in the UAV flight logs, especially for the azimuth angles. Although there were a few instances of significant deviations, the majority of the measurements were closely aligned with the angles calculated from the UAV flight logs, validating the effectiveness of the DF system for UAV direction estimation.

4.4. Analysis of the Real-Time Angle Measurements During the UAV's Flight Phase

The DF system enables real-time measurement of the azimuth and elevation angles during the UAV's flight. To assess the system's performance, real-time measurements were compared with the angles calculated from the UAV flight logs in two flight experiments. The real-time measurement values are obtained by averaging all of the direction-finding results in one second after removing outliers. The identification of outliers is based on a MAD criterion, defined as data points differing by more than three times the median deviation. For comparison, the angles per second are calculated from the UAV flight logs. The results are presented in Figures 10 and 11.



(a) Comparison of the real-time azimuth angle measurement results from the first flight.



(b) Comparison of the real-time azimuth angle measurement results from the second flight.

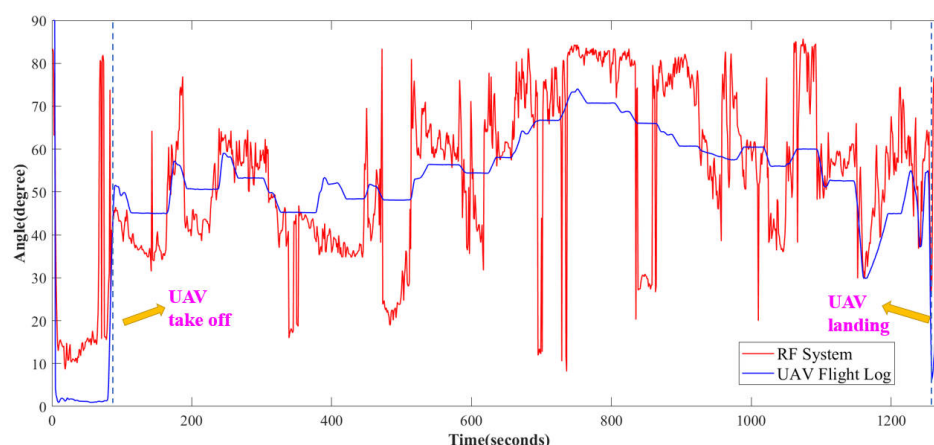
Figure 10. Comparison of real-time azimuth angle measurements from two flight experiments with the corresponding values from the UAV flight logs.

As shown in Figure 10, although deviations were observed between the azimuth angles measured in real time and the angles calculated from the UAV flight logs, the overall trend remained consistent after the removal of the outlier. The system exhibited small measurement errors, indicating excellent accuracy.

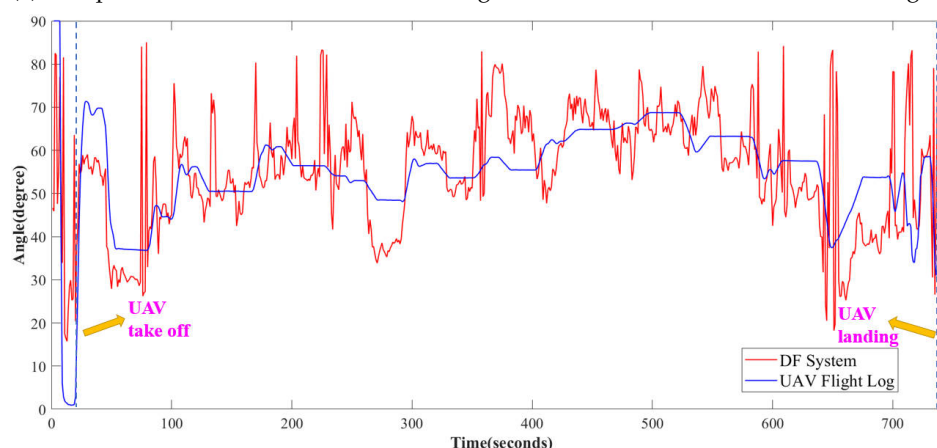
Figure 10a shows that there is a period of significant error between 800 and 900 s. By correlating this with the synchronously recorded elevation test results, this time corresponds to the 11th test point in the UAV hover test, and the error corresponds to a specific angle. Considering the suddenness and persistence of this error, the influence of burst signal interference in the environment should be excluded, as it should be the existence of obstacles in this direction that leads to serious multipath effects, resulting in serious deviations in the direction-finding results, and when the angle of the UAV changes,

the direction-finding accuracy reverts to its normal state. In addition, during the second flight, the measured azimuth angles were slightly higher, which was caused by deviations in the calibration during system initialization. However, the error remained within an acceptable range (about 10°).

It is also worth noting that significant measurement errors occurred at the beginning and end of both flights. These periods corresponded to the UAV's takeoff and landing phases, where the UAV was within the near-field region of the DF system. Consequently, these conditions resulted in larger measurement errors. As the UAV moved farther from the takeoff point into the far-field region, the system's measurement accuracy improved.



(a) Comparison of the real-time elevation angle measurement results from the first flight.



(b) Comparison of the real-time elevation angle measurement results from the first flight.

Figure 11. Comparison of real-time elevation angle measurements from two flight experiments with the corresponding values from the UAV flight log.

Figure 11 illustrates the real-time elevation angle measurements. Compared to the azimuth angles, the elevation angle measurements exhibited larger errors and fluctuations. This behavior is consistent with the inherent limitations of the UCA, which has reduced sensitivity in its elevation angle measurements as it nears the plane of the UCA antenna.

Despite these challenges, the DF system successfully captured the overall trend in the elevation angle changes, as indicated by the comparison with the values calculated from the flight logs. The measurement errors in the second flight were notably smaller and less fluctuating, reflecting the positive impact of the system's calibration process. Similar to the azimuth measurements, larger errors were observed during the takeoff and landing stages due to the UAV's proximity to the antenna array, highlighting the influence of near-field conditions on DF accuracy.

In summary, the analysis of the real-time angle measurements during the UAV's flight phase demonstrates the DF system's capability to effectively track the UAV's direction in dynamic flight scenarios. The azimuth measurements showed excellent accuracy, with minimal deviations compared to the UAV flight log results. The second flight, though it was slightly affected by hardware calibration deviations, still yielded acceptable results. The elevation angle measurements showed larger fluctuations, primarily due to the UAV's limitations and environmental factors, but the system successfully tracked the changes in the elevation angles. The measurement errors observed during the takeoff and landing phases underscore the challenges posed by near-field conditions.

Overall, the DF system provides reliable real-time UAV direction estimations, particularly for azimuth angles, and offers valuable insights for further improvements in the system's calibration and design to enhance the accuracy of elevation angle measurements.

4.5. Error Sources and Discussion

The observed error in direction estimation arises from three primary factors:

(1) The sensitivity of the elevation angle estimations of a UCA demonstrates distinct angular dependencies: when the elevation angle approaches the broadside direction ($\theta \rightarrow 0^\circ$), the phase difference between the antennas diminishes as $\sin \theta \rightarrow 0$, resulting in a widened main lobe beamwidth and reduced distinguishability between signals, thereby degrading the angular resolution and sensitivity. However, near the array plane ($\theta \rightarrow 90^\circ$), despite the phase difference reaching its maximum magnitude ($\sin \theta \rightarrow 1$), the inherent array symmetry introduces azimuthal ambiguities, often causing main lobe splitting or beamwidth expansion in the azimuthal dimension, which further deteriorates the accuracy of elevation estimation.

(2) Environmental interference, such as multipath reflections from nearby obstacles and random RF noise in outdoor settings, randomly distorts the coherence of the received signal waveforms, particularly during dynamic flight segments where rapid positional changes exacerbate these effects.

(3) While the WAA enhanced the computational efficiency by optimizing the MUSIC spectrum search process, its heuristic nature occasionally prioritized local optima over global peaks, leading to suboptimal angle estimations in complex signal environments.

These factors collectively highlight the need for future improvements, such as optimizing the antenna array structure to mitigate elevation angle ambiguities, incorporating advanced calibration techniques to reduce the errors caused by manual operation, and refining the WAA's exploration–exploitation balance to enhance the global convergence in spectral optimization.

5. Conclusions

This paper presents a UAV DF system utilizing radio frequency signals, specifically employing a UCA combined with the MUSIC algorithm and the WAA. This system addresses the critical challenges of UAV monitoring by using RF signals for reliable, real-time direction estimation. By optimizing the MUSIC algorithm with the WAA, this system significantly reduces the computational complexity and enhances the accuracy of both azimuth and elevation angle estimations.

The experimental results show that the proposed system can effectively estimate the azimuth and elevation angles of a UAV. Excluding outliers, the maximum error for the azimuth was 19.7° , while the minimum error was 0.1° , and the average azimuth measurement error was 7.0° . For the elevation angles, the maximum error was 23.5° , the minimum error was 0.2° , and the average error was 7.7° . These results were validated through real-time measurement data, confirming that the system could accurately track

UAVs in dynamic environments. These results show that the system is capable of providing reliable DF information, especially in the area of azimuth measurement, which is crucial for dynamic UAV monitoring and security applications.

Despite the system's overall effectiveness, limitations of the UCA, particularly in the elevation angle measurements, were observed. These limitations arise due to the geometry of the UCA, which reduces its sensitivity as it approaches the plane of the UCA. Furthermore, environmental factors, including multipath effects and RF interference, impacted the measurement accuracy. These results suggest that future work should focus on optimizing the array design, incorporating advanced calibration techniques, and refining the WAA's exploration–exploitation strategy to improve the performance in complex signal environments.

In conclusion, the proposed DF system offers a practical and cost-effective solution for UAV monitoring in low-altitude security applications, and the combination of the MUSIC algorithm and the WAA presents a powerful approach to UAV direction estimation. In view of the limitations in the direction-finding stability and accuracy caused by signal interference and multipath effects, future research will focus on further optimizing the algorithm to enhance its stability and anti-interference capabilities, as well as investigating more effective calibration methods. This will ensure robust support for the deployment and application of the system.

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